9. If A and B are independent events with $P(A) = P(B)$; and $P(A \cap B) = a$; then $P(B)$ is

(a) $2a$ (b) $\sqrt{a}$ (c) $\frac{a}{2}$ (d) a^2

10. If A and B are two events with P(A/B)= 0.3, P(B/A) = 0.2. thenP(A) is

(a) $\frac{3}{10}$ (b) $\frac{7}{10}$ (c) $\frac{6}{7}$ (d) $\frac{1}{7}$

II. Fill in the blanks:

11. The probability of the entire sample space is _____

12. On throwing the single die, then the event of getting odd number or even number are _____ event.

13. Probability of getting a Monday in a week is _____

14. If A_1 and A_2 are independent events, then $P(A_1 \cup A_2) = p(A_1) +$ _____

15. If $A \subset B$, then $P(A)$___$P(B)$.

16. If probability for the occurrence of an event is 1, then the event is known as _____ event.

17. If $P(A) = 0$, then A is called _____ event.

18. Axiomatic approach to probability was proposed by _____

19. If A and B are two events with $P(A \cup B) = \frac{10}{15}$, then $P(\bar{A} \cap \bar{B})$ = _____

20. The conditional probability P(A/B) can be calculated using $P(B)$,if _____

III Very Short Answer Questions:

21. If E_1 and E_2 are two mutually exclusive events and Given that $P(E_2) = 0.5$ and $P(E_1 \cup E_2) = 0.7$ then find $P(E_1)$.

22. Find the probability that a leap year selected at random will contain 53 Fridays?

23. Two coins are tossed simultaneously. What is the probability of getting exactly two heads?

24. If $P(A \cap B) = 0.3$, $P(B) = 0.7$ find the value of $P(A/B)$

25. A box contains 5 red and 4 white marbles. Two marbles are drawn successively from the box without replacement and it is noted that the second one is white. What is the probability that the first is also white?